AKHIL R

Computer Science & Engineering Student | Backend Development | AI/ML
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Summary

Final-year Computer Science student with internship experience in backend development, cybersecurity, and applied AI, along with four national-level hackathon finals and a published research paper. Passionate about building scalable backend systems and delivering reliable software solutions within collaborative engineering teams.

Technical Skills

Languages: Java, Python, SQL
Backend: Django, Django REST Framework, Flask
Machine Learning: PyTorch, Scikit-learn, Pandas, NumPy, Computer Vision, YOLO
Databases: MySQL, SQLite
Tools: Git, Swagger, Docker

Experience

Backend Developer Intern June 2025 - July 2025
VR Della Infotech *Trichy, India*

- Built an e-commerce backend with accounts, catalog, cart, and orders, delivering 12+ REST endpoints using Django REST Framework and MySQL.
- Secured role-based access across all endpoints by implementing token-based authentication with endpoint testing.
- Enabled reliable checkout logic with full CRUD coverage by designing cart-to-order business logic.

Cybersecurity Intern April 2025 - June 2025
Cothon Solutions *Remote*

- Blocked unauthorized data movement by monitoring file system, email, USB, and clipboard in a Python/PyQt5/Watchdog DLP app.
- Enforced six-department policies via role-based access, anomaly detection, and encrypted storage.
- Validated reliability with a 96.97% test pass rate (32/33) and 12 flagged alerts using a PyQt5 reporting dashboard.

AI & ML Intern February 2025 - March 2025
Edunet Foundation *Remote*

- Built an MRI-based brain tumor detector with binary classification and confidence scoring by fine-tuning ResNet18 via transfer learning.
- Standardized 224x224 RGB scan preprocessing by training and validating the model on the BraTS dataset.
- Delivered real-time confidence-scored predictions per scan by serving the model through a Flask REST endpoint.

Projects

Marine Biodiversity Intelligence System | *Python, DNABERT, HDBSCAN, UMAP, Flask*

- Built a Flask dashboard embedding DNA sequences into 768-dim vectors via DNABERT 6-mer tokenization, classifying species using a classifier trained on NCBI barcode data.
- Grouped sequences into high-purity species clusters via UMAP and HDBSCAN, flagging novel/unmatched sequences.
- Built a batch pipeline computing Shannon, Simpson, and Evenness indices with CSV export and FASTA uploads.

PADCOM - AI-Powered Medical Diagnostic Platform | *Python, Flask, PyTorch, MONAI*

- Built a multi-condition diagnostic platform detecting fractures, brain tumors, and pneumonia via Flask REST endpoints.
- Unified three architectures - ResNet50, ResNet18, and a MONAI DenseNet model - into one imaging pipeline.
- Improved diagnostic reviewability by building a grayscale-to-color visualization pipeline for every scan.

Education

B.E. Computer Science & Engineering 2023 - 2027
Kongunadu College of Engineering and Technology, Trichy CGPA: 7.3
Higher Secondary Certificate 2023
Chellamal Matric Hr. Sec. School Percentage: 77.3%

Publications & Achievements

- **Published** "VoiceGuard - AI-Based Voice Authenticity Detection System" in IJSRET Journal, Vol. 12, Issue 2 (2026)
- **Finalist** - ANRF AISEHack 2026, IIIT Hyderabad
- **Finalist** - IISF 2025 Hackathon, CSIO, Haryana
- **Top Team** - Health Hackathon, VIT Bhopal
- **Finalist** - HackDay Pondy (0x.Day & SMVEC)
- Contributed to KNCET's official **Idealab** website - idealab.kongunadu.ac.in